

1-zero and BCFW shift

up to cyclic permutations a 1-zero is the imposition of the $n - 3$ constraints

$$C_{13} = C_{14} = \cdots = C_{1,n-1} = 0 \qquad \text{on the kinematic data.}$$

Since $C_{ij} = X_{ij} + X_{i+1,j+1} - X_{i,j+1} - X_{i+1,j}$

the constraints can be equivalently be written as

$$\left\{ \begin{array}{l} X_{13} + X_{24} - X_{14} - \textcolor{brown}{X_{23}}_0 = 0 \\ X_{14} + X_{25} - X_{15} - X_{24} = 0 \\ \vdots \\ X_{1,n-1} + X_{2n} - \textcolor{brown}{X_{1n}}_{=0} - X_{2,n-1} = 0 \end{array} \right. \iff \left\{ \begin{array}{l} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ \vdots \\ X_{2n} = X_{1n} - X_{13} = -X_{13} \end{array} \right.$$

so the 1-zero fully fixes X_{2j} once $X_{1\delta}$ is specified, and does not affect $X_{i\delta}$ if $i, \delta \notin \{1, 2\}$.

the corresponding BCFW shift is the change

$$\left\{ \begin{array}{l} P_2 \rightarrow P_2 + zq \\ P_n \rightarrow P_n - zq \end{array} \right.$$

where z is a complex parameter and q satisfies

$$\left\{ \begin{array}{l} q^2 = 0 \\ q \cdot P_2 = q \cdot P_n = 0 \end{array} \right.$$

$$\searrow \text{ so that } \begin{array}{l} (P_2 + zq)^2 = 0 \\ (P_n - zq)^2 = 0 \end{array}$$

Let $X_{ij}(z)$ be the z -dependent planar invariant.

$X_{ij}(z)$ depends on z only if it includes exactly one of P_2, P_n in the sum, otherwise either there is no z or they cancel out.

$$\rightarrow \text{ the only } z\text{-dependent ones are } \quad X_{1j}(z) \quad j = 3, \dots, n-1 \\ \text{ and } \quad X_{2j}(z) \quad j = 4, \dots, n.$$

Let X_{ij}^∞ denote the coefficient of the largest power of z appearing in $X_{ij}(z)$.